

**SentryBand: A Wearable Real-Time Heat Stress and Toxic Gas Exposure Alert System for  
Marine Engine Room Personnel**

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## **SentryBand: A Wearable Real-Time Heat Stress and Toxic Gas Exposure Alert System for Marine Engine Room Personnel**

The SentryBand is a wrist-worn safety device designed for the engineers and oilers who stand watch inside a ship's engine room, one of the most physically demanding and hazardous workspaces onboard a vessel. It continuously tracks two things that current engine room routines rarely measure in real time: the wearer's own heat-strain level, through skin temperature and heart rate, and the surrounding concentration of toxic or combustible gases, such as carbon monoxide and hydrogen sulfide. When either reading approaches an unsafe threshold, the band vibrates and sounds an alarm on the wearer's wrist while simultaneously relaying the alert to a dashboard in the engine control room, giving both the crew member and the officer of the watch enough warning to respond before a heat-related collapse or a gas-exposure incident occurs.

The name "SentryBand" was chosen to reflect exactly what the device does and how it is worn. A sentry is a person or mechanism that stands watch and warns of danger before it escalates, which is precisely the function this wearable performs for its user, continuously watching over conditions the wearer cannot always sense on their own, since gas concentrations, for instance, can build up long before symptoms appear. "Band" is a direct, descriptive reference to its physical form as a wristband, making the title easy to picture and immediately understandable to both technical and non-technical readers. Together, the title captures the project's core purpose, a personal, wearable guardian for engine room crew, without resorting to overly technical jargon, while still signaling that this is a functional safety instrument rather than a lifestyle gadget.

### **Theoretical Background**

Ship engine rooms are consistently identified as one of the hottest and most physically taxing working environments on a vessel because of the combined effect of engine heat radiation, poor ventilation in enclosed compartments, and the physical exertion required to complete watchkeeping rounds (Layuk et al., 2022; Palella et al., 2016). Palella et al. (2016) identified the engine room as one of the ship compartments most consistently classified as a severe hot environment from a thermo-hygrometric standpoint, and noted that shipboard heat-stress management still commonly relies on heat-stress indices that are outdated or have not been rigorously validated, because little systematic monitoring data exists; they proposed adapting the SOBANE risk-management strategy, already used in land-based industry, to address this gap at sea. This finding is reinforced by Layuk et al. (2022), whose field study of

ship engine room crew found that shipboard temperature, noise, and vibration levels exceeded the applicable threshold limit values and were measurably associated with crew work fatigue. Oldenburg and Jensen (2019) similarly found that, among the occupational groups aboard container ships, engine room personnel carried a substantial physical workload, second only to deck ratings, while nautical officers, though physically less taxed, reported the highest mental stress and the lowest variability in heart rate of the three groups, underscoring that engine room duty imposes a distinct and measurable physical burden even where it is not the single most extreme occupational category onboard. A later field study by Oldenburg et al. (2020), conducted across six container ship voyages, found that engine room personnel were exposed to the highest average noise levels of any occupational group (96 dB) and that noise, vibration, and heat were the physical stressors seafarers most frequently linked to psychological stress during the voyage. Taken together, these studies establish that heat, noise, and physical strain in the engine room are not an occasional inconvenience but a recurring, cumulative, and largely unmonitored occupational hazard.

Beyond heat exposure, human fatigue compounds the safety risk of the engine room. In a simulated engine-room exercise, Symes et al. (2025) measured engine room operators' brain activity with functional near-infrared spectroscopy while participants worked through a scenario requiring them to identify and resolve a simulated fault, and found that fatigue significantly increased activation in the brain's prefrontal cortex during the task, indicating that a fatigued operator must expend more cognitive effort to catch and correct the same fault, direct physiological evidence that fatigue measurably taxes an engineer's capacity to notice and respond to developing problems. Chowdhury et al. (2024), reviewing maritime human-factors research published between 2010 and 2022, found that human error remains a persistent contributor to marine accidents and argued that existing mitigation strategies still center on seafarer training, competency-building, and procedural collaboration rather than on giving the at-risk crew member any real-time technological feedback about their own condition. This is a critical gap: the STCW safety-training framework that engine officers are certified under (International Maritime Organization, n.d.-b) equips them with the knowledge to recognize hazards, but training alone cannot compensate for a human being's reduced physiological capacity to notice a slow, cumulative rise in heat strain or gas concentration in the moment it is happening.

Toxic and oxygen-deficient atmospheres represent an even more acute hazard tied to the engine room and its adjacent enclosed spaces, such as tanks, bilges, and void spaces accessed during maintenance. According to industry safety data compiled by SAFETY4SEA

(2025), roughly 350 seafarers and shore-based workers have lost their lives to asphyxiation while inside enclosed spaces aboard ships since 1996, with 43 recorded incidents between 2022 and the start of 2025 alone accounting for 70 deaths, a trend that has not meaningfully slowed despite the International Maritime Organization's 2011 enclosed-space entry resolution. Gard (2025), a major protection-and-indemnity insurer, echoed this concern in its updated guidance on enclosed-space entry, emphasizing that structured risk assessment before entry remains inconsistently applied across vessel types, particularly for cargo-related hazards that release gas gradually rather than in an obvious, detectable burst. While fixed multi-gas detection systems and portable connected detectors such as Honeywell's BW Ultra (Honeywell International Inc., 2018) already exist for confined-space entry teams, these tools are typically deployed only at the moment of a planned enclosed-space entry; they are not designed to be worn continuously by engine room watchkeepers during routine duties, when unplanned leaks from aging pipework or fuel systems are just as capable of causing harm.

This monitoring gap is widest on smaller, domestic vessels. The International Maritime Organization has repeatedly flagged that fatal ferry incidents are heavily concentrated on domestic routes rather than international ones, with domestic voyages accounting for roughly 95% of such casualties, prompting it to adopt Model Regulations on Domestic Ferry Safety in 2022 and to fund regional safety-improvement projects across Southeast Asia (International Maritime Organization, n.d.-a). These smaller vessels rarely carry the class-notated, fixed physiological and gas-monitoring infrastructure found on larger, internationally trading ships, largely because such systems are costly to install and certify. The result is a two-tier safety landscape: well-resourced ocean-going vessels increasingly adopt automation and monitoring, while domestic ferries and small cargo or passenger craft, which carry the bulk of casualties, continue to rely almost entirely on manual vigilance and periodic engine room rounds.

Fortunately, the underlying sensing technology needed to close this gap is already mature and inexpensive outside the maritime sector. Sadeghi et al. (2025) designed and prototyped a low-cost, AI-assisted wearable capable of continuously tracking vital signs and issuing intelligent alerts, demonstrating that consumer-grade microcontrollers and biosensors can achieve clinically meaningful accuracy in a compact form factor. Ponce et al. (2025) validated a commercial IoT and LoRaWAN-based wearable monitoring system for isolated workers inside a real steel plant, demonstrating via network-coverage and radio-interference trials that industrial-grade smartwatches can reliably relay real-time vital-sign alerts to safety supervisors across a large industrial facility, a deployment model directly transferable to an engine control room dashboard. On the environmental side, the U.S. National Institute for

Occupational Safety and Health (2026) maintains a validated heat-safety index and companion application that translates ambient conditions into actionable exposure guidance, providing a ready-made, evidence-based threshold framework rather than requiring a new one to be developed from scratch. What none of these efforts has done, however, is combine physiological heat-strain sensing and toxic or combustible gas sensing into a single low-cost wearable purpose-built for the specific thermal, mechanical, and atmospheric conditions of a ship's engine room.

Synthesizing these bodies of research reveals a clear and specific gap: engine room crew face a documented, compounding combination of heat strain, fatigue, and toxic-gas risk (Layuk et al., 2022; Palella et al., 2016; Symes et al., 2025; SAFETY4SEA, 2025) that is least monitored on precisely the smaller, domestic vessels that account for most maritime casualties (International Maritime Organization, n.d.-a); meanwhile, the wearable sensing and IoT-alerting technology required to address this gap already exists in mature, low-cost form in adjacent industries (Sadeghi et al., 2025; Ponce et al., 2025; Honeywell International Inc., 2018) but has not been adapted to this specific shipboard context. SentryBand is proposed as that adaptation: a single wrist-worn device that fuses physiological heat-strain sensing with toxic and combustible gas sensing, uses the NIOSH heat-exposure framework and established gas-detection thresholds as its safety logic, and delivers alerts both to the wearer and to a supervising officer, at a cost and complexity appropriate for the small and medium-sized vessels that need it most.

## **Project Description**

### **Project Objectives**

This project aims to achieve the following objectives:

- To design a compact, wrist-worn device that continuously monitors physiological heat-strain indicators, namely skin temperature and heart rate, together with ambient toxic and combustible gas concentrations, namely carbon monoxide, hydrogen sulfide, and flammable vapors, within the engine room environment.
- To develop a two-tier, real-time alert mechanism that combines an immediate on-wrist vibration, LED, and buzzer warning with a wireless notification relayed to a dashboard in the engine control room, so that both the wearer and the officer of the watch are informed before conditions reach a hazardous threshold.
- To increase the affordability and accessibility of personal condition-monitoring technology for engine room crew aboard small and medium-sized domestic vessels that typically cannot afford the fixed, class-approved gas-detection and monitoring systems installed on larger, internationally trading ships.

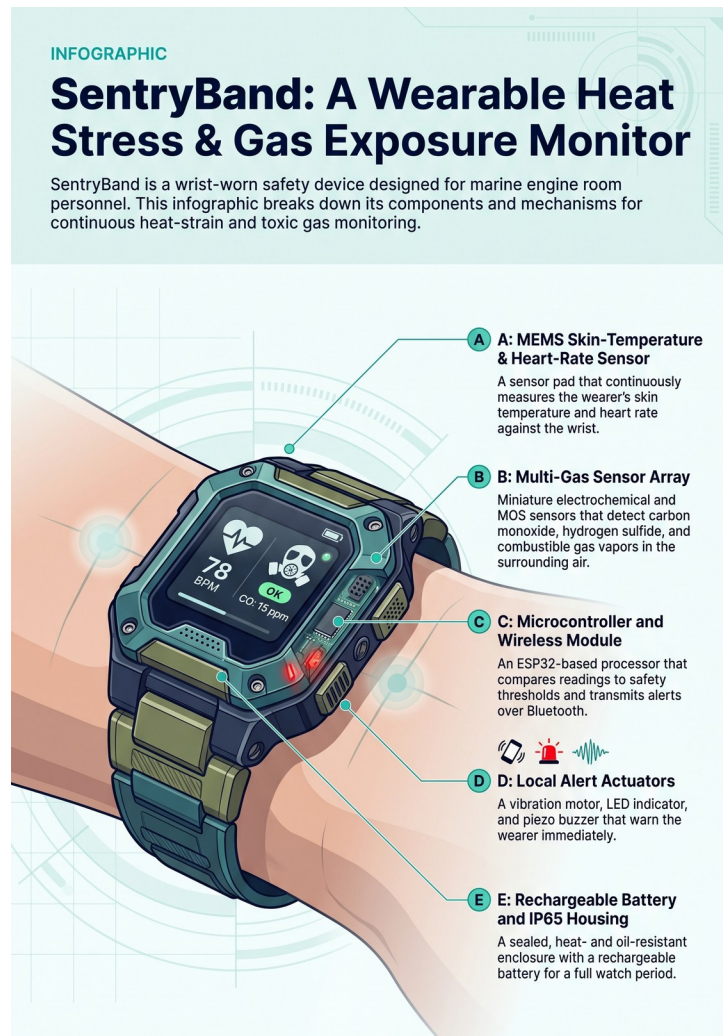
### **Project Features**

The project includes the following key features:

- Dual-Sensing Core – Combines a skin-contact temperature and heart-rate sensor with a miniature multi-gas sensor array in a single wrist-worn unit, allowing it to monitor the wearer's own physiological state and the surrounding air at the same time.
- Tiered Alert System – Pairs an immediate, connectivity-independent local alarm (vibration, LED, buzzer) with a wireless Bluetooth relay to a control-room dashboard, ensuring the wearer is warned instantly while supervisors gain situational awareness of the whole crew.
- Rugged, Low-Power Wearable Housing – Encloses the electronics in an IP65-rated, heat- and oil-resistant casing with a rechargeable battery sized for a full multi-hour watch, so the device can be retrofitted onto any crew member's routine without modifying shipboard systems.

## Mechanism and Process Descriptions

### Mechanism Descriptions



**Note:** Image generated through Higgsfield AI

#### 1. MEMS Skin-Temperature and Heart-Rate Sensor Module

This component sits against the inside of the wrist and continuously measures the wearer's skin temperature alongside heart rate using photoplethysmography. It functions as the device's physiological heat-strain indicator, converting subtle rises in body temperature and heart rate into an early warning signal before the wearer consciously feels overheated.

#### 2. Multi-Gas Sensor Array

This component combines electrochemical sensors tuned to carbon monoxide and hydrogen sulfide with a metal-oxide semiconductor sensor for combustible vapors. It functions as the device's environmental sensing layer, sampling the air immediately

around the wearer to detect toxic or flammable gas buildup that may originate from leaking fuel lines, exhaust manifolds, or poorly ventilated bilge spaces.

### **3. Microcontroller and Wireless Communication Unit**

This is an ESP32-based processing board that reads data from both sensor modules, compares each reading against pre-set heat-strain and gas-exposure thresholds, and manages both the local alarm and the Bluetooth Low Energy connection to the control-room dashboard. It functions as the processing core of the device, deciding when a reading has crossed from normal to hazardous.

### **4. Local Alert Actuators (Vibration Motor, LED Indicator, and Piezo Buzzer)**

This set of components delivers an immediate, multi-sensory warning directly on the wearer's wrist. It functions as the device's first line of defense, ensuring the crew member is alerted even in loud engine rooms or in areas without wireless signal.

### **5. Rechargeable Battery and IP65-Rated Enclosure**

This component consists of a compact lithium-ion battery and a sealed, heat- and oil-resistant housing that protects the internal electronics. It functions as the device's power source and physical shield, allowing it to survive a full watch in the heat, oil mist, and vibration typical of an engine room.

## **Process Description**

The operation of SentryBand follows these main steps:

### **Step 1: Activation and Calibration**

At the start of a watch, the crew member powers on the device, which records a short baseline reading of the wearer's resting skin temperature and heart rate and allows the gas sensors to stabilize, a standard warm-up period for electrochemical and metal-oxide-type sensors.

### **Step 2: Continuous Sensing**

Throughout the watch, the MEMS sensor module and the multi-gas sensor array sample readings at fixed short intervals, sending each reading to the microcontroller for processing.

### **Step 3: Threshold Comparison**

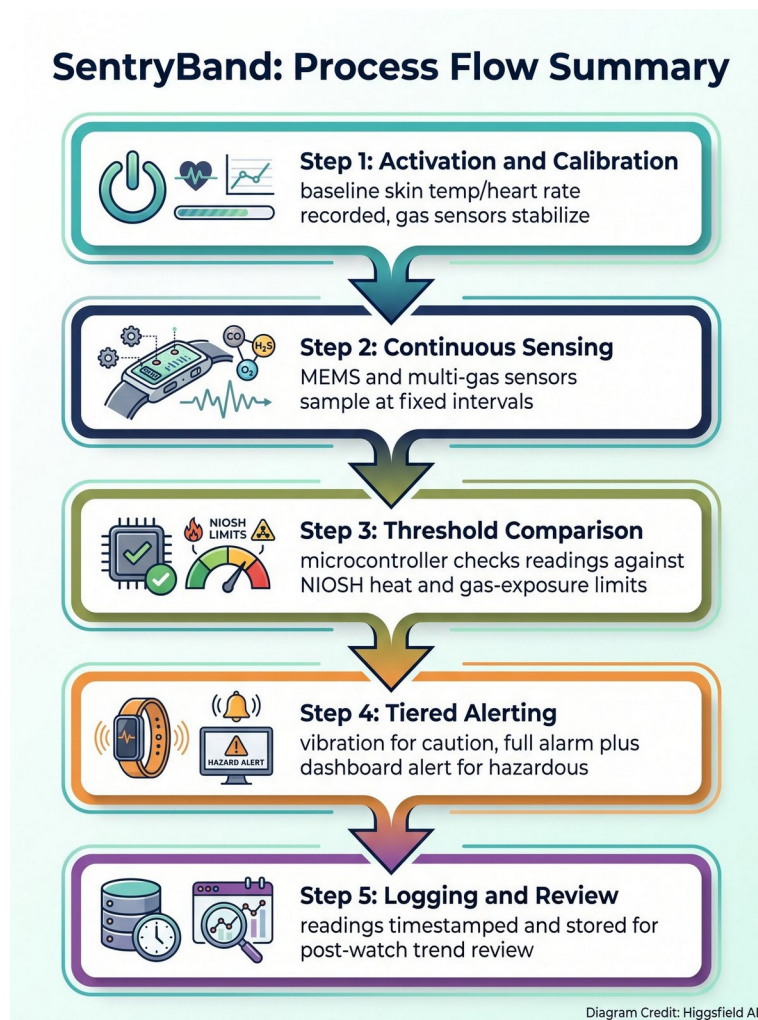
The microcontroller compares each incoming reading against pre-set safety thresholds derived from the NIOSH heat-exposure framework and established occupational gas-exposure limits, classifying the current condition as normal, caution, or hazardous.

#### Step 4: Tiered Alerting

If a reading enters the caution range, the device issues a brief vibration alert to prompt the wearer to check their surroundings or take a break; if a reading crosses into the hazardous range, the device triggers its full local alarm, buzzer and flashing LED, and simultaneously transmits an alert to the engine control room dashboard.

#### Step 5: Logging and Review

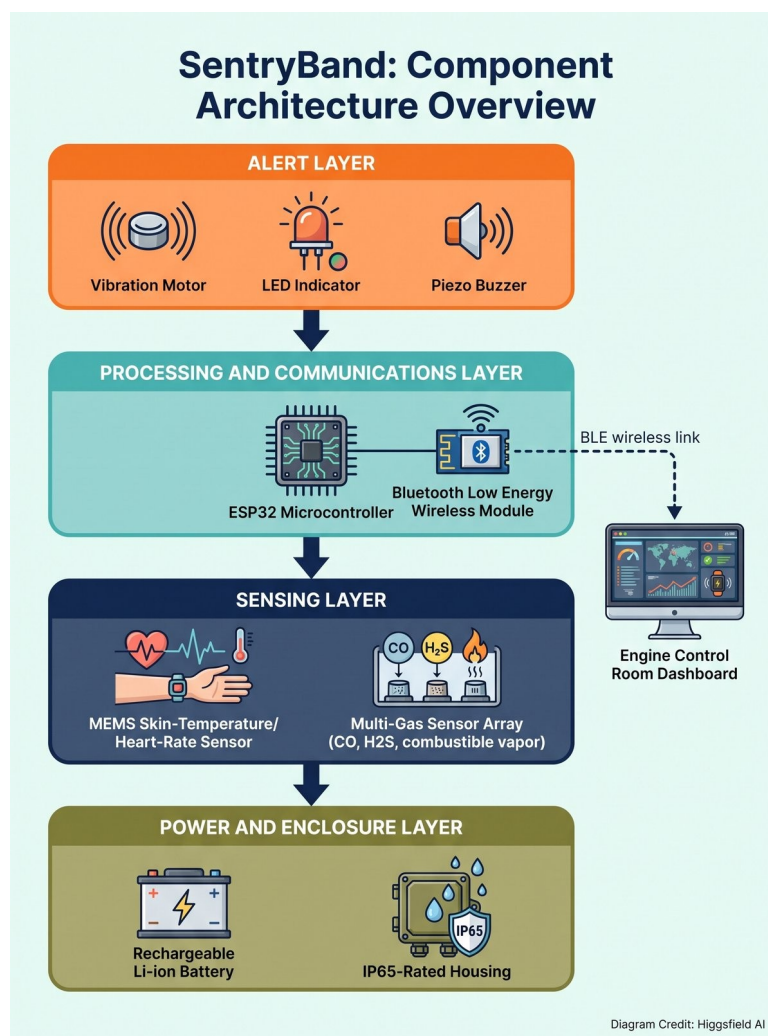
All readings and alerts are timestamped and stored both on the device and on the dashboard, allowing the chief engineer to review exposure trends after the watch and identify recurring hot spots or gas-prone areas within the engine room.



## Assembly and Testing

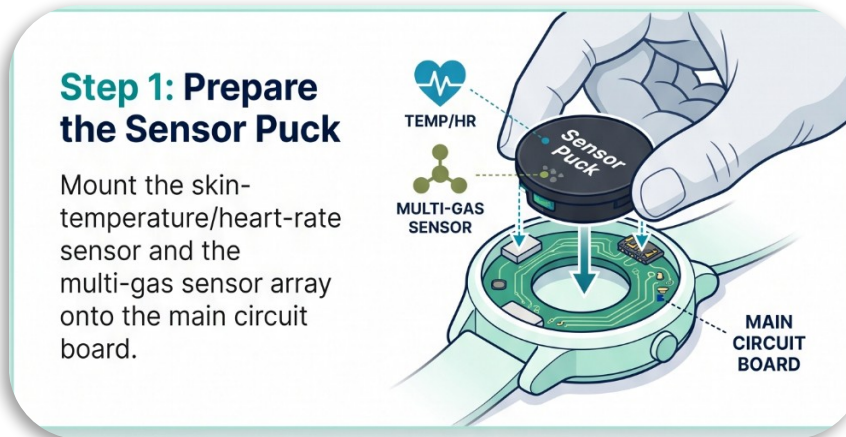
### Assembly

The assembly of SentryBand follows a streamlined process that brings the sensing, processing, and alert components together into a single wearable unit, integrated across the four functional layers shown in the figure below. The wireless relay between the wearable and the control-room dashboard uses Bluetooth Low Energy rather than Wi-Fi: BLE's connection and duty-cycling model has been measured to consume markedly less energy per byte transferred than comparable low-power wireless standards in continuous-use embedded devices (Siekkinen et al., 2012), letting the radio sleep between periodic sensor updates rather than maintaining the continuous, higher-power association that Wi-Fi requires, a critical consideration for a device that must run for a full multi-hour watch on a small rechargeable battery. The assembly of the project follows these main steps:



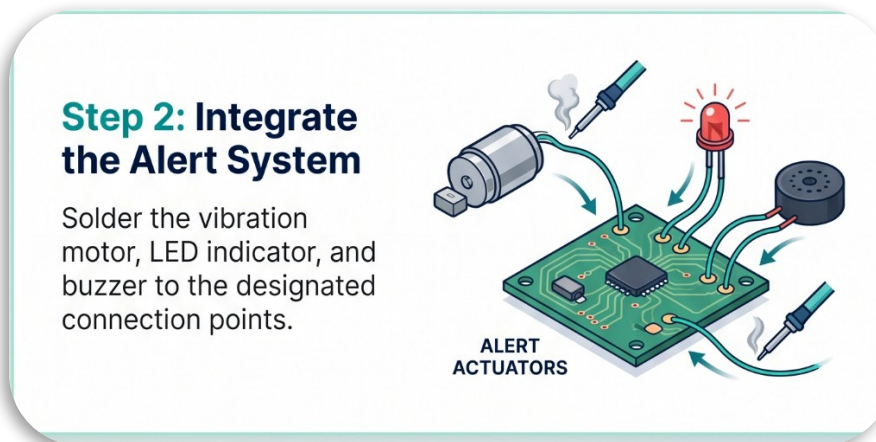
### 1. Prepare the Sensor Puck

Mount the MEMS skin-temperature/heart-rate sensor and the multi-gas sensor array onto the main circuit board, positioning the skin-contact sensor so it faces outward through the housing window.



### 2. Integrate the Alert Actuators

Solder the vibration motor, LED indicator, and piezo buzzer to their designated connection points on the circuit board, then connect the ESP32 microcontroller and confirm each actuator responds to a test signal.

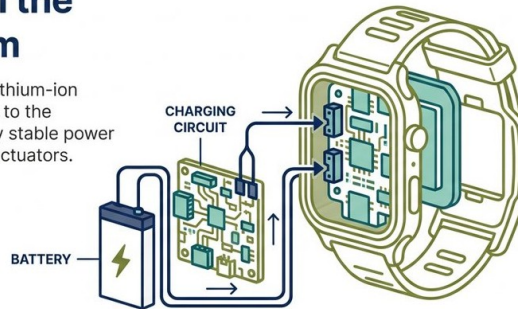


### 3. Install the Power System

Connect the rechargeable lithium-ion battery and charging circuit to the assembled board, and verify stable power delivery to all sensors and actuators.

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Connect the rechargeable lithium-ion battery and charging circuit to the assembled board, and verify stable power delivery to all sensors and actuators.



## 4. Seal and Enclose

Fit the assembled electronics into the IP65-rated wristband housing, aligning the skin-contact sensor window and gas-sensor vents, then seal the casing.

### Step 3: Seal and Attach the Wearable Strap

Seal the assembled electronics into the water- and oil-resistant housing, then attach the adjustable wristband strap.



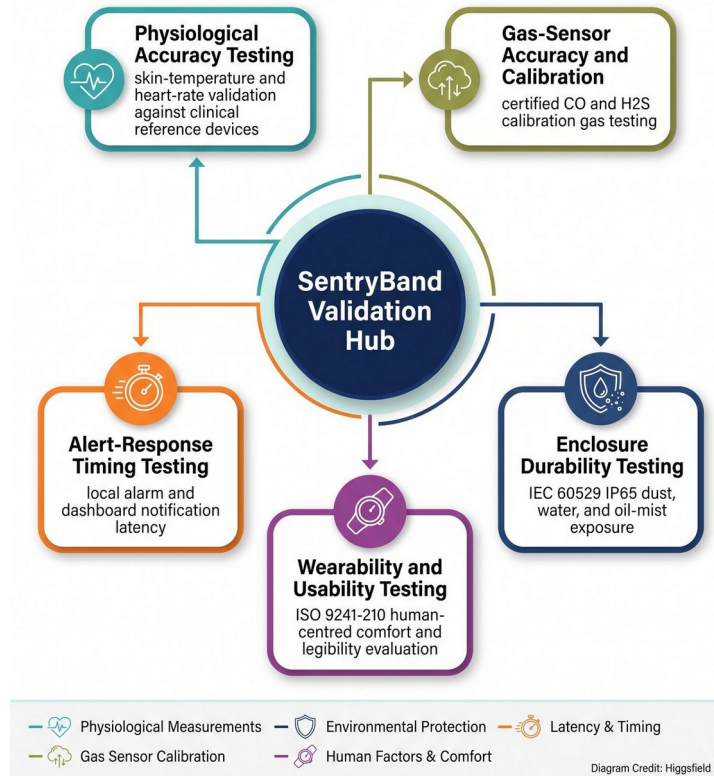
## 5. Attach the Wearable Strap

Attach the adjustable wristband strap to the sealed housing, sized to fit securely over a work glove or bare wrist.

## Testing

Since physical prototyping and shipboard trials were not feasible within the scope of this report, research-based testing protocols were developed by examining how comparable physiological and gas-sensing devices are validated in industrial and consumer settings, and by drawing on established international standards for each dimension of the device's performance. The figure below summarizes the five testing dimensions used to validate SentryBand, each addressing a distinct aspect of the device's design.

# SentryBand: Testing Framework Overview



## Physiological Sensor Accuracy Testing

Drawing on the validation approach used by Sadeghi et al. (2025) for their AI-powered vital-signs wearable, this test would compare SentryBand's skin-temperature and heart-rate readings against a reference clinical thermometer and pulse oximeter under simulated engine room heat conditions, confirming that the wearable's readings remain within a clinically acceptable margin of error before the device is trusted to trigger a heat-strain alert.

## Multi-Gas Sensor Accuracy and Calibration Testing

Following the calibration and bump-testing workflow built into connected portable gas detectors such as the Honeywell BW Ultra, whose docking station automatically produces calibration and compliance paperwork (Honeywell International Inc., 2018), SentryBand's multi-gas sensor array would undergo a comparable scheduled calibration-gas test to confirm reading accuracy before each deployment and at scheduled intervals during service.

### **Enclosure Durability Testing**

An enclosure durability test would combine the IP65 ingress-protection method defined by the International Electrotechnical Commission (n.d.) in IEC 60529, which exposes the sealed housing to dust and low-pressure water jets, with an additional oil-mist exposure specific to the engine room environment, since IP65 alone does not certify resistance to the oil mist and hydrocarbon vapor typical of that space; together these would confirm the electronics remain protected across a simulated multi-month service period.

### **Alert-Response Timing Testing**

An alert-response test would measure the time elapsed between a reading crossing the hazardous threshold and both the local alarm and the control-room dashboard notification, benchmarked against the exposure-time guidance in the NIOSH (2026) heat-safety framework, to confirm the wearer and officer of the watch are warned early enough to respond safely. This test would also confirm that the Bluetooth Low Energy link chosen for the wireless relay, selected for its markedly lower energy cost per byte transferred than comparable low-power wireless standards in continuous-use embedded devices (Siekkinen et al., 2012), still delivers the alert within the required response window even under the electromagnetic interference typical of an engine room.

### **Wearability and Usability Testing**

Following ISO 9241-210's human-centred design process for interactive systems (International Organization for Standardization, 2019), a wearability and usability test would evaluate whether engine room personnel can read and act on SentryBand's alerts without confusion or excessive cognitive load while wearing standard personal protective equipment such as gloves and coveralls. This test would assess comfort and freedom of movement across a full watch, the legibility of the local LED and vibration cues in high-noise, low-visibility conditions, and whether wearers can correctly distinguish a caution-level alert from a hazardous-level alert without consulting the control-room dashboard.

Together, these five testing dimensions would confirm not only that SentryBand's individual sensors function accurately, but that the complete device performs reliably, comfortably, and communicates clearly under the specific thermal, mechanical, and atmospheric conditions of a ship's engine room.

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## Appendix

### Generative AI Usage Reporting Sheet for Image Generation and Revision Phase

I acknowledge the use of generative AI to assist in generating images/revising my technical report. While many revisions were made manually, this sheet documents only the instances where generative AI tools were used. The table below provides a detailed explanation of how AI was applied in revising each section of my report.

| Generative AI Used  | Page # - Section                                                                            | Prompt Used                                                                                                                                                                                                                                                                                                                         | Generated Content                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Higgsfield          | Mechanism Descriptions section                                                              | Infographic-style prompt requesting a labeled device-anatomy diagram of a wrist-worn safety wearable, with callouts A-E identifying the skin-temperature/heart-rate sensor, multi-gas sensor array, microcontroller/wireless module, local alert actuators, and battery/IP65 enclosure, in a clean teal-and-navy infographic style. | A labeled infographic illustration of the SentryBand device on a wrist, with callouts A-E matching the five components described in the Mechanism Descriptions section.                                                                                                                                                                                                                             |
| Higgsfield & Runway | Assembly section (Steps 1-5 illustrations)                                                  | Infographic-style prompt requesting a 3-step assembly guide showing the sensor puck being mounted, the alert actuators being soldered, and the electronics being sealed into the wristband housing, in the same infographic style as the mechanism diagram.                                                                         | A 4-card assembly infographic (cropped into three separate step illustrations) matching the five written Assembly steps.                                                                                                                                                                                                                                                                            |
| Higgsfield          | Process Description section                                                                 | Infographic-style prompt requesting a 5-step circular process-flow diagram showing activation/calibration, continuous sensing, threshold comparison, tiered alerting, and logging/review, in the same teal-and-navy infographic style as the mechanism diagram.                                                                     | A 5-step circular process-flow infographic matching the five written Process Description steps.                                                                                                                                                                                                                                                                                                     |
| Higgsfield          | Assembly section (architecture overview)                                                    | Infographic-style prompt requesting a layered architecture diagram showing the four functional layers of the wearable (sensing, processing/communication, alerting, power/enclosure) and how they connect, in the same infographic style as the other diagrams.                                                                     | A 4-layer architecture infographic showing how the sensing, processing, alerting, and power components connect.                                                                                                                                                                                                                                                                                     |
| Higgsfield          | Testing section                                                                             | Infographic-style prompt requesting a 5-panel summary of the testing dimensions (physiological sensor accuracy, multi-gas sensor accuracy, enclosure durability, alert-response timing, wearability/usability), in the same infographic style as the other diagrams.                                                                | A 5-panel testing-framework infographic matching the five named testing subsections.                                                                                                                                                                                                                                                                                                                |
| Claude (Anthropic)  | Testing section                                                                             | Asked the AI to reorganize the single-paragraph Testing section into clearly labeled subsections, one per test type, for readability.                                                                                                                                                                                               | Before: one paragraph listing testing ideas generally → After: a short introductory paragraph followed by 5 named subsections ("Physiological Sensor Accuracy Testing," "Multi-Gas Sensor Accuracy and Calibration Testing," "Enclosure Durability Testing," "Alert-Response Timing Testing," "Wearability and Usability Testing"), each with its own subheading, plus a closing summary paragraph. |
| Claude (Anthropic)  | Figure captions throughout (Mechanism Descriptions, Process Description, Assembly, Testing) | Asked the AI to standardize the wording of the figure captions under each diagram so they read consistently throughout the report.                                                                                                                                                                                                  | Before: "(Image generation: AI image-generation tool)" → After: a consistent short attribution line under each of the four full-width figures.                                                                                                                                                                                                                                                      |

